1. User_Type Table

Stores user type information.

Column Name	Data Type	Constraints
user_type_id	INT	PRIMARY KEY, AUTO_INCREMENT
type_name	VARCHAR(50)	NOT NULL, UNIQUE

2. Users Table

Stores user information.

Column Name	Data Type	Constraints
user_id	INT	PRIMARY KEY, AUTO_INCREMENT
fname	VARCHAR(50)	NOT NULL
lname	VARCHAR(50)	NOT NULL
email	VARCHAR(100)	NOT NULL, UNIQUE
phoneno	VARCHAR(15)	NOT NULL
user_type_id	INT	NOT NULL, FOREIGN KEY REFERENCES user_type(user_type_id)
registration_date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

3. Login Table

Stores user login information.

Column Name	Data Type	Constraints
login_id	INT	PRIMARY KEY, AUTO_INCREMENT
user_id	INT	NOT NULL, FOREIGN KEY REFERENCES users(user_id)
email	VARCHAR(100)	NOT NULL, UNIQUE
password	VARCHAR(255)	NOT NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

4. Categories Table

Stores product categories.

Column Name	Data Type	Constraints
category_id	INT	PRIMARY KEY, AUTO_INCREMENT
category_name	VARCHAR(100)	NOT NULL, UNIQUE
description	TEXT	NULL

5. Products Table

Stores product details.

Column Name	Data Type	Constraints
product_id	INT	PRIMARY KEY, AUTO_INCREMENT
product_name	VARCHAR(100)	NOT NULL, UNIQUE
category_id	INT	NOT NULL, FOREIGN KEY REFERENCES categories(category_id)
price	DECIMAL(10, 2)	NOT NULL, CHECK (price > 0)
stock_quantity	INT	NOT NULL, CHECK (stock_quantity >= 0)
sunlight_requirement	ENUM('Full Sun', 'Partial Shade', 'Full Shade')	NOT NULL
water_need	ENUM('Low', 'Medium', 'High')	NOT NULL
climate_compatibility	ENUM('Tropical', 'Subtropical', 'Temperate', 'Arid')	NOT NULL
growth_rate	ENUM('Slow', 'Moderate', 'Fast')	NOT NULL
soil_type	ENUM('Sandy', 'Loamy', 'Clay', 'Peaty', 'Silty')	NOT NULL
flowering_season	ENUM('Spring', 'Summer', 'Autumn', 'Winter')	NOT NULL
height_range	ENUM('Short', 'Medium', 'Tall')	NOT NULL
description	TEXT	NULL
image_url	VARCHAR(255)	NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

6. Cart Table

Stores user cart details.

Column Name	Data Type	Constraints
cart_id	INT	PRIMARY KEY, AUTO_INCREMENT
user_id	INT	NOT NULL, FOREIGN KEY REFERENCES users(user_id)
product_id	INT	NOT NULL, FOREIGN KEY REFERENCES products(product_id)
quantity	INT	NOT NULL, CHECK (quantity > 0)
added_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

7. Orders Table

Stores order details.

Column Name	Data Type	Constraints
order_id	INT	PRIMARY KEY, AUTO_INCREMENT
user_id	INT	NOT NULL, FOREIGN KEY REFERENCES users(user_id)
order_date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
total_amount	DECIMAL(10, 2)	NOT NULL
shipping_address	VARCHAR(255)	NOT NULL
payment_method	ENUM('UPI', 'Card', 'Netbanking')	NOT NULL
status	ENUM('Pending', 'Processed', 'Shipped', 'Delivered', 'Cancelled')	NOT NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

8. Order_Details Table

Stores individual product details within an order.

Column Name	Data Type	Constraints
order_detail_id	INT	PRIMARY KEY, AUTO_INCREMENT
order_id	INT	NOT NULL, FOREIGN KEY REFERENCES orders(order_id)
product_id	INT	NOT NULL, FOREIGN KEY REFERENCES products(product_id)
quantity	INT	NOT NULL, CHECK (quantity > 0)
price	DECIMAL(10, 2)	NOT NULL

9. Delivery Table

Stores delivery information.

Column Name	Data Type	Constraints
delivery_id	INT	PRIMARY KEY, AUTO_INCREMENT
order_id	INT	NOT NULL, FOREIGN KEY REFERENCES orders(order_id)
delivery_person_id	INT	NOT NULL, FOREIGN KEY REFERENCES users(user_id)
delivery_date	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
status	ENUM('Pending', 'In Transit', 'Delivered')	NOT NULL

10. Reviews Table

Stores product reviews by users.

Column Name	Data Type	Constraints
review_id	INT	PRIMARY KEY, AUTO_INCREMENT
user_id	INT	NOT NULL, FOREIGN KEY REFERENCES users(user_id)
product_id	INT	NOT NULL, FOREIGN KEY REFERENCES products(product_id)
rating	INT	NOT NULL, CHECK (rating >= 1 AND rating <= 5)
comment	TEXT	NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

11. Plant_Care Table

Stores plant care guidelines.

Column Name	Data Type	Constraints
plant_care_id	INT	PRIMARY KEY, AUTO_INCREMENT
product_id	INT	NOT NULL, FOREIGN KEY REFERENCES products(product_id)
planting_method	ENUM('Raised bed', 'Pit')	NOT NULL
pit_size	VARCHAR(10)	NULL
canopy_width	VARCHAR(10)	NULL
canopy_height	VARCHAR(10)	NULL
season	ENUM('January-February', 'March-April', 'May-June', 'July-August', 'September-October', 'November-December')	NOT NULL
ideal_location	ENUM('Low land', 'Plains')	NOT NULL
ideal_temperature	VARCHAR(20)	NOT NULL
ideal_rainfall	VARCHAR(20)	NOT NULL
ideal_humidity	VARCHAR(20)	NOT NULL
soil_ph	VARCHAR(10)	NOT NULL
soil_type	ENUM('Well drained', 'Sandy loam', 'Laterite')	NOT NULL
elevation	VARCHAR(20)	NOT NULL
shade_tolerance	ENUM('Very low', 'Low', 'Moderate', 'High')	NOT NULL
pest_and_disease	TEXT	NULL

12. Registered_Plants Table

Stores details of plants registered for monitoring.

Column Name	Data Type	Constraints
registered_plant_id	INT	PRIMARY KEY, AUTO_INCREMENT
user_id	INT	NOT NULL, FOREIGN KEY REFERENCES users(user_id)
product_id	INT	FOREIGN KEY REFERENCES products(product_id)
plant_name	VARCHAR(100)	NOT NULL
purchase_date	DATE	NOT NULL
monitoring_start_date	DATE	NOT NULL